Unsupervised Decomposition of 3D Shapes into Expressive and Editable Extruded Profile Primitives

CHUNYI SUN, Australian National University, Australia
JUNLIN HAN, University of Oxford, United Kingdom
RUNJIA LI, University of Oxford, United Kingdom
WEIJIAN DENG, Australian National University, Australia
DYLAN CAMPBELL, Australian National University, Australia
STEPHEN GOULD, Australian National University, Australia



Fig. 1. **Decomposition of 3D shapes into extruded profile primitives.** In each group, the input 3D representation (voxel grid) is to the left, the primitives are in the middle, and the reconstructed 3D shape with path curves is on the right. Our 3D2EP method generates compact, interpretable representations with a small set of primitives, providing a structural abstraction that supports flexible editing while closely aligning with the original geometry.

Transforming 3D shapes into representations that support part-level editing, flexible redesign, and efficient compression is vital for asset customization, content creation, and optimization in digital design. Despite its importance, achieving a representation that balances expressivity, editability, compactness, and interpretability remains a challenge. We introduce 3D2EP, a novel method for 3D shape decomposition that represents objects as a collection of differentiable, parametric primitives. Given a 3D shape represented by a voxel grid, 3D2EP decomposes this into a set of primitive parts, each generated by extruding a scaled 2D profile along a 3D curve, with the requisite components being predicted in a feedforward manner. That is, each primitive is constrained to have a single cross-section profile, up to scale. This enables the primitives to adapt to the data, capturing the geometry with precision but without excess degrees-of-freedom that would stymie editability.

Extensive evaluations highlight 3D2EP's ability to reconstruct complex shapes with a compact and interpretable representation, emphasizing its suitability for a wide range of 3D modeling applications.

Authors' Contact Information: Chunyi Sun, Australian National University, CANBERRA, Australia, chunyi.sun@anu.edu.au; Junlin Han, University of Oxford, London, United Kingdom, junlinhan@meta.com; Runjia Li, University of Oxford, Oxford, United Kingdom, runjiali@outlook.com; Weijian Deng, Australian National University, CANBERRA, Australia, weijian.deng@anu.edu.au; Dylan Campbell, Australian National University, CANBERRA, Australia, dylan.campbell@anu.edu.au; Stephen Gould, Australian National University, CANBERRA, Australia, stephen.gould@anu.edu.au.



This work is licensed under a Creative Commons Attribution 4.0 International License.
SIGGRAPH Conference Papers '25, Vancouver, BC, Canada
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1540-2/25/08
https://doi.org/10.1145/3721238.3730704

CCS Concepts: • **Computing methodologies** → **Parametric curve and surface models**; **Shape representations**; *Reconstruction.*

Additional Key Words and Phrases: Unsupervised 3D Decomposition, Editable Primitives, Shape Abstraction

ACM Reference Format:
Chunyi Sun, Junlin Han, Runjia Li, Weijian Deng, Dylan Campbell, and Stephen Gould. 2025. Unsupervised Decomposition of 3D Shapes into Expressive and Editable Extruded Profile Primitives. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH Conference Papers '25), August 10–14, 2025, Vancouver, BC, Canada.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3721238.3730704

1 Introduction

Recent advances in 3D shape generation from scans and single-view images have greatly improved the reconstruction of realistic 3D models [Han et al. 2025, 2024; Hong et al. 2023; Li et al. 2024; Lin et al. 2023; Poole et al. 2022; Raj et al. 2023; Sun et al. 2023], enabling applications in digital design, gaming, and augmented reality. However, many generated assets lack features such as part-level manipulation, efficient compression, and flexible redesign, limiting their adaptability to tasks that require detailed control and customization. Implicit representations, such as neural radiance fields (NeRFs) [Mildenhall et al. 2021] and signed distance functions (SDFs) [Park et al. 2019], excel in capturing intricate geometries but prioritize reconstruction quality over editability. Their non-locality makes precise manipulation challenging, reducing suitability for flexible workflows. Explicit representations, like Gaussian splatting [Kerbl et al. 2023], are also unsuitable, as editing conglomerations of Gaussians is non-trivial.

In contrast, primitive-based representations offer explicit control using simple shapes like cubes, spheres, or basic extrusions. Yet, representing fine details often requires numerous primitives, leading to inefficient, overly complex models impractical for many tasks.

We propose 3D2EP, an approach that balances expressivity and editability by decomposing a 3D shape into extruded profile primitives. 3D2EP fits primitives directly to each input 3D shape with only 2D part segmentation masks as supervision. The decomposition of 3D shapes into extruded profile primitives is shown in Fig. 1. Each primitive is constructed by extruding a scaled 2D profile along a 3D curve, accommodating a wide range of shape variations. Specifically, 3D2EP processes the voxelized input shape through a neural network that predicts a set of curves and corresponding scale parameters. These predictions are used to compute bounding boxes along each curve, which segment the input voxel grid and allow 2D occupancy grid profiles to be computed. This approach directly fits the profiles to the data, with the free parameters only controlling the extrusion path and scaling. Constraining the degrees-of-freedom in this way facilitates editing and manipulation, without losing any expressivity with respect to the 2D profile. The path curves, 2D occupancy grid profiles, and scale parameters are then integrated to reconstruct the final 3D shape, achieving a compact and interpretable representation.

The anisotropic scaling along the extrusion path enhances the expressiveness of the representation by allowing the size and aspect ratio of the cross-section profile to vary. Our profile prediction is highly accurate, enabling the representation of complex shapes with intricate details. This flexibility surpasses the limitations of traditional extruded primitives [Li et al. 2023a; Paschalidou et al. 2019; Yang and Chen 2021], which lack such adaptability. Combined with a fully differentiable formulation and efficient conversion to occupancy volumes, our method supports fast optimization and seamless integration of 3D occupancy optimization and mask rendering guidance. By leveraging segmentation masks, we achieve semantically meaningful primitive decomposition, ensuring reasonable and intuitive shape partitioning.

3D2EP performs test-time, per-instance optimization without any pre-training. This avoids the cost and ambiguity of manual annotations required by supervised methods, enabling scalable and generalizable shape decomposition.

Extensive validation on the GC-Object [Sawdayee et al. 2023; Zhou et al. 2015] and Quadrupeds [Tulsiani et al. 2017a; Xu et al. 2023] datasets demonstrates the ability of 3D2EP to represent complex 3D shapes with high fidelity while maintaining a compact and interpretable representation. 3D2EP supports practical applications in 3D shape editing, interpolation, and parameter reduction by combining curve extrusion, scaling, and learned 2D occupancy maps to capture intricate geometric details. Our key contributions are:

(1) a method for decomposing 3D shapes into parametric primitives, enabling flexible 3D asset editing and redesign;
(2) a differentiable extruded profile primitive representation with adaptive scaling, offering an expressive and concise shape abstraction; and
(3) applications of the above to shape editing and interpolation.

2 Related Work

2.1 Implicit 3D Representations

Implicit 3D representations have gained popularity for their ability to represent complex shapes. Early methods like OccNet [Tong et al. 2023], IM-Net [Chen and Zhang 2019], and DeepSDF [Park et al. 2019] pioneered coordinate-based neural implicits, mapping 3D coordinates to shape descriptors. However, these focused on global features, leading to over-smoothed results lacking fine details. Subsequent work incorporated local features using voxels [Chabra et al. 2020; Chibane et al. 2020; Jiang et al. 2020; Peng et al. 2020; Tang et al. 2021], image grids [Li and Zhang 2021; Saito et al. 2019], or surface points [Boulch and Marlet 2022; Giebenhain and Goldlücke 2021; Huang et al. 2023; Williams et al. 2022], to condition implicit networks. Recent NeRF [Barron et al. 2021, 2023; Li et al. 2023b; Lyu et al. 2024; Mildenhall et al. 2021; Müller et al. 2022; Verbin et al. 2022; Wang et al. 2021, 2023; Zhang et al. 2020] series further advanced implicit representations, excelling at view synthesis. While good at capturing details, these methods often produce dense, uninterpretable representations. This lack of structure makes part-level editing or semantic understanding difficult.

Our work, unlike existing implicit methods, introduces a structured representation through differentiable parametric primitives. This bridges the gap between implicit representations' expressiveness and the need for explicit control, enabling detailed shape editing while maintaining high fidelity.

2.2 Explicit 3D Representations

Explicit 3D representations provide interpretability and control by directly encoding geometric information. Traditional methods, such as meshes and point clouds, are widely used in computer graphics and offer detailed representations. However, they struggle with topological changes and require significant storage for complex shapes. Recent methods such as SHRED [Jones et al. 2022] bypass explicit primitive modeling by directly predicting part-aware point cloud patches, achieving fine-grained reconstructions but lacking structured, controllable representations for editing or semantic understanding. Parametric primitives offer a more compact alternative by representing shapes with a small set of parameters. Beyond basic primitives, advanced methods such as superquadrics [Barr 1981; Paschalidou et al. 2020, 2019], cuboids [Kluger et al. 2021; Niu et al. 2018; Smirnov et al. 2020; Sun et al. 2019; Tulsiani et al. 2017b; Zou et al. 2017], anisotropic Gaussians [Genova et al. 2019], and convexes [Deng et al. 2020] have been proposed. Despite their compactness, they often struggle to represent complex shapes with intricate details and may require a large number of primitives to achieve high fidelity, leading to increased model complexity and optimization challenges.

Our method builds upon explicit representations by introducing differentiability to parametric primitives. This enables efficient end-to-end optimization, achieving high-fidelity reconstructions while maintaining fine-grained control over individual parts of the shape.

2.3 3D Reverse Engineering

3D reverse engineering aims to reconstruct structured, interpretable 3D models from raw data. Early methods use simple primitives

such as cuboids [Tulsiani et al. 2017b; Zou et al. 2017] and superquadrics [Paschalidou et al. 2019] to represent shapes compactly, but struggle with complex or organic structures due to limited geometric expressiveness.

To overcome this, recent approaches [Li et al. 2019; Liu et al. 2024; Sharma et al. 2020; Tulsiani et al. 2017b; Yan et al. 2021; Zhao et al. 2024; Zou et al. 2017] decompose shapes into structured parametric parts, aiming to balance fidelity and interpretability. SweepNet [Zhao et al. 2024] predicts occupancy from path curves and simplified profiles, but its reliance on indirect supervision and coarse representations often leads to overlapping primitives, slower convergence, and reduced editability. Without semantic guidance, part decompositions can also be inconsistent or less meaningful.

In contrast, our 3D2EP method introduces a more expressive primitive representation based on extruded profiles along spatial paths. It supports efficient optimization, incorporates semantic cues during decomposition, and produces smooth, compact, and editable outputs suited for practical modeling tasks.

3 Decomposing a 3D Shape with 3D2EP

The 3D2EP method decomposes a 3D shape into simple but expressive parametric primitives. The key observation is that object parts can usually be well-approximated by extrusions of a scaled 2D profile along a curve in 3D. As such, we constrain the primitives to each have a single, potentially non-convex profile that is extruded along a smooth curve.

3.1 The 3D2EP Primitive Representation

Given an input shape represented by an occupancy volume $X \in \mathbb{R}^{V\times V\times V}$ with resolution V, each 3D2EP primitive is parameterized by (1) a curve $c\colon [0,1] \to \mathbb{R}^3$, defined by N B-spline control points, and (2) a scaling function $s\colon [0,1] \to \mathbb{R}^2$, defined by P linear-spline control points. This fully defines the occupancy volume O for the primitive, as shall be outlined in the following sections. Hence, each primitive has $3N + 2P$ parameters, as well as the parameters associated with its occupancy grid profile, which are computed from the input occupancy volume.

3.1.1 Coordinate System Transformation. Let $\{\mathbf{c}_i\}_{i=0}^{P-1}$ denote P linearly-sampled points along the curve c, where $\mathbf{c}_i = c\big(i/(P-1)\big)$, and let $\{\mathbf{s}_i\}_{i=0}^{P-1}$ represent the corresponding scaling factors from s. These samples, as depicted in Fig. 2 (left), divide the curve into $(P-1)$ segments and illustrate the relationship between the curve points and scaling factors. For each segment, we define a coordinate system transformation that maps a point in 3D to the segment's canonical frame.

Specifically, for the ith segment, we define its associated translation, scaling factor, and height as

$$\mathbf{t}_i = \frac{\mathbf{c}_i + \mathbf{c}_{i+1}}{2}, \quad \mathbf{f}_i = \frac{\mathbf{s}_i + \mathbf{s}_{i+1}}{2}, \quad \text{and } h_i = \|\mathbf{c}_{i+1} - \mathbf{c}_i\|, \tag{1}$$

respectively, and the rotation matrix as

$$R_i = \begin{bmatrix}\mathbf{b}_i & \mathbf{n}_i & \mathbf{v}_i\end{bmatrix}, \tag{2}$$

where the tangent, binormal and normal vectors are given by

$$\mathbf{v}_i = \frac{\mathbf{c}_{i+1} - \mathbf{c}_i}{\|\mathbf{c}_{i+1} - \mathbf{c}_i\|}, \quad \mathbf{b}_i = \frac{\mathbf{r}\times\mathbf{v}_i}{\|\mathbf{r}\times\mathbf{v}_i\|}, \quad \text{and } \mathbf{n}_i = \frac{\mathbf{v}_i\times\mathbf{b}_i}{\|\mathbf{v}_i\times\mathbf{b}_i\|}, \tag{3}$$

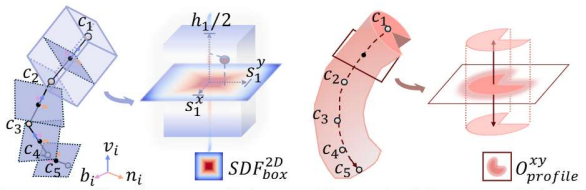


Fig. 2. **Curve-Based Extrusion Process.** The left side illustrates sampling five points along a curve c, dividing it into 4 segments. Each segment is defined by its midpoint $\mathbf{t}_i = \frac{\mathbf{c}_i+\mathbf{c}_{i+1}}{2}$, scaling factor $\mathbf{s}_i$, and height h_i, with a local orientation ($R_i = [b_i, n_i, v_i]$) determining the extrusion direction. The right side illustrates the visualization of the extruded profile occupancy map in 3D, where the bounding box defines a cutting plane to extract the profile occupancy map from the provided occupancy volume.

for a reference vector $\mathbf{r} = (0,0,1)$. Then, a 3D point $\mathbf{x} = (x,y,z)$ in the global coordinate frame is mapped to the segment's canonical frame according to

$$\mathbf{x}'_i = \operatorname{diag}\big([\mathbf{f}_i,\ 1/h_i]\big)\big(R_i\,\mathbf{x} - \mathbf{t}_i\big). \tag{4}$$

3.1.2 Computing a Coarse, Bounding Occupancy Volume. To obtain a coarse version of the primitive, we first model the 2D profile as a square, which is anisotropically scaled and extruded along the curve, thereby producing a simple curved 3D bounding shape that can be later refined. This process is visually summarized in Fig. 2, where the left part illustrates the SDF of a single 3D box extrude from the 2D box SDF.

In the local frame of each segment, a 2D signed distance function (SDF) models the cross-section as a square with half-width w, and is given by

$$\mathrm{SDF}^{2D}_{\mathrm{box}}(x',y') = \|(x',y')\| - w. \tag{5}$$

Extruding along the normalized z'-axis corresponds to

$$\mathrm{SDF}^{3D}_{\mathrm{box}}(\mathbf{x}') = \max\big(\mathrm{SDF}^{2D}_{\mathrm{box}}(x',y'),\ |z'| - 0.5\big), \tag{6}$$

where 0.5 is the normalized half-height. In the global coordinate system, this corresponds to the SDF of a cuboid located along that curve segment. We then define the associated occupancy volume as

$$O_{\mathrm{box}}(\mathbf{x}') = \sigma\big(-\alpha\,\mathrm{SDF}^{3D}_{\mathrm{box}}(\mathbf{x}')\big), \tag{7}$$

where σ is the sigmoid function and α controls boundary sharpness.

3.1.3 Computing a Fine-grained Occupancy Volume. So far we have defined a square profile that, once scaled and extruded, provides a coarse, blocky shape for the primitive. In this section, we show how to estimate a fine-grained 2D profile from the data. Specifically, we propose a slicing-based approach that samples and averages cross-sectional profiles from the high-resolution occupancy volume X^+ which is a detailed version of the original low-resolution input X. As illustrated in Fig. 3 (bottom-right), this process produces a refined profile. Extruding this profile leads to a much higher fidelity reconstruction of the part.

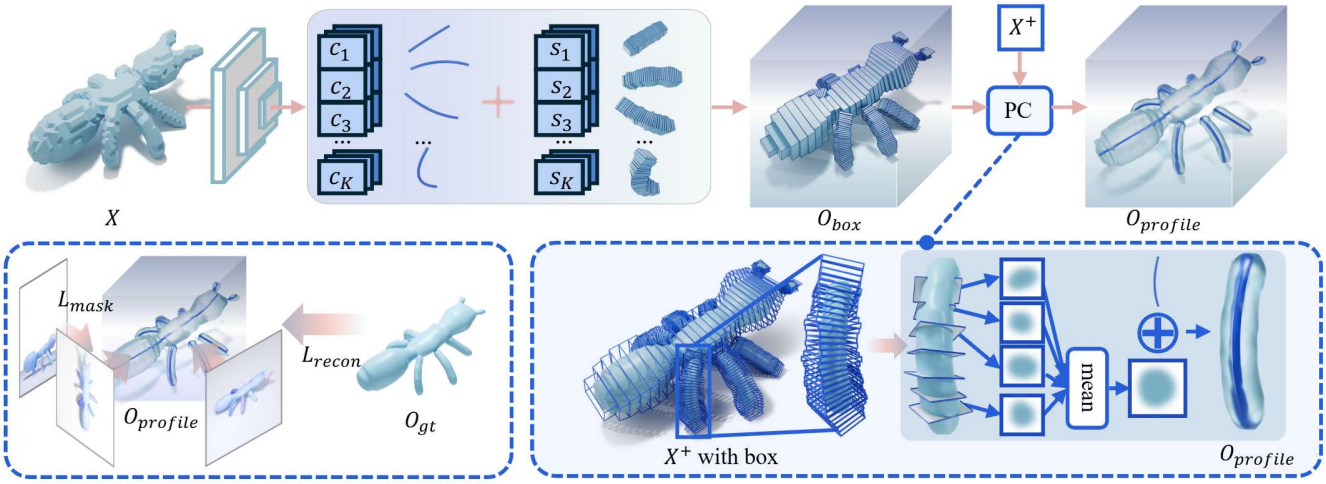


Fig. 3. **Overview of the 3D2EP method for decomposing 3D shapes into extruded profile primitives.** The top diagram illustrates the full pipeline: the voxelized input shape X is processed by the network to predict path curves and corresponding scale parameters. These predictions are used to compute bounding boxes, which are used to slice the high-resolution input shape X^+ in order to compute 2D occupancy grid profiles for each primitive. The path curves, 2D profiles, and scales are then integrated to reconstruct the final 3D shape. The bottom-left panel shows the optimization process, combining a 3D reconstruction loss and mask rendering guidance to refine the results. The bottom-right panel demonstrates the process for computing per-primitive cross-section profiles: slicing planes are sampled along the curve and the corresponding occupancy grids are averaged to obtain a single cross-section for the primitive. Together with the curve and scaling parameters, this forms a very compact and interpretable shape representation.

Extracting 2D Profiles via Slicing. At each sample point $\mathbf{t}_i$, we define a slicing plane using the curve's binormal $\mathbf{b}_i$ and normal $\mathbf{n}_i$ vectors, previously defined. A bounding box on this plane is discretized at resolution $R_{\text{slice}} \times R_{\text{slice}}$. For $(u_1, u_2) \in [-0.5, 0.5]^2$, we have the normalized plane

$$\mathbf{p}_i(u_1, u_2) = \frac{2}{V^+ - 1} (\mathbf{t}_i + u_1\, \mathbf{b}_i + u_2\, \mathbf{n}_i) - 1, \tag{8}$$

in the coordinates of data structure X^+. Trilinearly interpolating occupancy values in X^+ at the 3D coordinates corresponding to a set of uniformly-spaced 2D points on the plane allows us to compute the occupancy grid of the 2D slice, given by

$$O_{\text{profile},i} = \text{Interpolate}\big(X^+,\ \{\mathbf{p}_i(u_1, u_2)\}\big) \tag{9}$$

for $u_1 \in [0, R_{\text{slice}} - 1]/(R_{\text{slice}} - 1) - 0.5$ and u_2 likewise.

Averaging the Slices. We then average the profile slices to form a single cross-sectional profile for the primitive, given by

$$\bar{O}_{\text{profile}} = \frac{1}{P-1} \sum_{i=0}^{P-2} O_{\text{profile},i}. \tag{10}$$

From this, the occupancy of a query point $\mathbf{x}'$ in the canonical coordinate system can be computed via

$$O^{xy}_{\text{profile}}(x', y') = \text{Interpolate}\big(\bar{O}_{\text{profile}}, (x', y')\big). \tag{11}$$

Height-dependent Occupancy. To prevent the extrusion from extending beyond each segment's bounding box, we include a soft cutoff term that depends on the z coordinate, given by

$$O^{z}_{\text{profile}}(z') = \sigma\big(\lambda\, (|z'| - 0.5)\big), \tag{12}$$

where λ controls sharpness and 0.5 is the normalized half-height.

3D Profile Occupancy. Multiplying the cross-sectional and height-based components yields

$$O_{\text{profile}}(\mathbf{x}') = O^{xy}_{\text{profile}}(x', y')\, O^{z}_{\text{profile}}(z'), \tag{13}$$

a smooth, high-fidelity volumetric representation.

3.1.4 Combining 3D Occupancies. So far, for each primitive, we have an analytic but coarse 3D representation O_{box} and a fine-grained 3D representation O_{profile} that leverages the input occupancy volume data to recover finer details. These are multiplied to obtain the final 3D occupancy $O = O_{\text{box}}\, O_{\text{profile}}$, which has the effect of removing any nonzero occupancy outside the bounding boxes induced by O_{box}.

Finally, we aggregate the occupancy across all K primitives, each with $(P-1)$ segments:

$$\hat{O}(\mathbf{x}') = \sum_{j=1}^{K} \sum_{i=0}^{P-2} O_{ij}(\mathbf{x}'), \tag{14}$$

where j indexes over the primitives. This yields a complete volumetric reconstruction of the input shape.

3.2 Network Architecture

In this section, we outline how the parameters described in Section 3.1 are predicted by our model in a feedforward manner. As shown in Fig. 3, our approach factors a voxel grid $X \in \mathbb{R}^{V \times V \times V}$ into K differentiable geometric primitives, each parameterized by a 3D

curve and an associated scaling function. To do so, we implement a voxel encoder and a parameter decoder.

3.2.1 Voxel Encoder. A 3D convolutional encoder first processes the voxel grid X into a latent representation $\mathbf{z} \in \mathbb{R}^D$, capturing essential geometric features of the input shape:

$$\mathbf{z} = f(X; \boldsymbol{\theta}), \tag{15}$$

where $\boldsymbol{\theta}$ encompasses the parameters of the encoder's 3D convolutional and fully-connected layers.

3.2.2 Parameter Decoder. The decoder predicts parameters for K primitives from $\mathbf{z}$. Each primitive is defined by N B-spline control points given by the set $\boldsymbol{Q}_j = \{\mathbf{q}_{jk} \in \mathbb{R}^3\}_{k=0}^{N-1}$ and P linear-spline control points $\boldsymbol{S}_j = \{\mathbf{s}_{ji} \in \mathbb{R}^2\}_{i=0}^{P-1}$. Formally,

$$\{\boldsymbol{Q}_j,\ \boldsymbol{S}_j\} = g(\mathbf{z}; \boldsymbol{\phi}) \tag{16}$$

with network parameters $\boldsymbol{\phi}$. The decoder thus provides all required parameters for each primitive: B-spline and scaling control points.

3.3 Optimization

Optimization of the network parameters is guided by a multi-term loss, given by

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{recon}} + \lambda_2 \mathcal{L}_{\text{overlap}} + \lambda_3 \mathcal{L}_{\text{axis}} + \lambda_4 \mathcal{L}_{\text{mask}}, \tag{17}$$

where the coefficients $\lambda_{1,\dots,4}$ balance the reconstruction accuracy, overlap minimization, axis alignment, and mask consistency.

Reconstruction Loss ($\mathcal{L}_{recon}$). This term encourages both the combined predicted occupancy $\hat{O}(\mathbf{x})$ and the box-based occupancy $O_{\text{box}}(\mathbf{x})$ to match the ground-truth occupancy $O_{\text{gt}}(\mathbf{x})$. For query points $\mathbf{x}$ in a sampling set $\mathcal{X}$, the loss is given by

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{\mathbf{x}\sim\mathcal{X}}\Big(\big\|\hat{O}(\mathbf{x}) - O_{\text{gt}}(\mathbf{x})\big\|_2^2 + \big\|O_{\text{box}}(\mathbf{x}) - O_{\text{gt}}(\mathbf{x})\big\|_2^2\Big). \tag{18}$$

Overlap Loss ($\mathcal{L}_{overlap}$). Redundant coverage by multiple primitives is penalized by comparing the sum of per-primitive occupancies to a fixed threshold 1. The loss is given by

$$\mathcal{L}_{\text{overlap}} = \mathbb{E}_{\mathbf{x}\sim\mathcal{X}}\Big[\max\Big(\sum_{j=1}^{K} O_j(\mathbf{x}) - 1,\ 0\Big)\Big]. \tag{19}$$

Axis Alignment Loss ($\mathcal{L}_{axis}$). Predicted curves are steered to align with points on the medial axes $\mathbf{m} \in \mathcal{M}$. For each point $\mathbf{m}$, the distance to the closest point $\mathbf{c}$ in the set of predicted curves C is computed, giving

$$\mathcal{L}_{\text{axis}} = \mathbb{E}_{\mathbf{m}\sim\mathcal{M}}\big[\min_{\mathbf{c}\in C} \text{dist}(\mathbf{m},\ \mathbf{c})\big]. \tag{20}$$

Mask Consistency Loss ($\mathcal{L}_{mask}$). A silhouette-based term enforces alignment between a rendered occupancy silhouette S_{rendered} and the ground-truth silhouette S_{gt}. Summing over image pixels $(h, w) \in \{1, \dots, H\} \times \{1, \dots, W\}$, the loss is

$$\mathcal{L}_{\text{mask}} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\delta\left(\sqrt{1 + \frac{\big(S_{\text{rendered}}(h,w) - S_{\text{gt}}(h,w)\big)^2}{\delta^2}} - 1\right), \tag{21}$$

where δ acts to smooth the gradients.

All four terms jointly guide the network toward accurate volumetric reconstruction, minimal overlap, faithful axis alignment, and coherent silhouettes.

The medial axes are computed as in Tagliasacchi et al. [2012]. For part segmentation, we densely render the 3D object from multiple views, randomly select one, and use the SAM automatic image segmentation model [Kirillov et al. 2023] to predict a mask. Following Qin et al. [2024], we hierarchically divide the SAM result into three levels and select subpart-level masks. This reference mask is then used to guide the SAM video segmentation network [Ravi et al. 2024], ensuring multiview-consistent masks.

4 Experiments

4.1 Implementation Details and Datasets

3D2EP is designed for efficient per-shape optimization. Each input shape is optimized independently using 2,500 iterations using the Adam optimizer [Zhang 2018]. The shape is decomposed into primitives, with the number of primitives determined by the mask count. For all experiments, we use $P = 32$ sampled points on the curve, $N = 3$ B-spline control points. The input voxel grid V for the network has a resolution of 64^3, while the cutting profile plane voxel grid V^+ is set to a higher resolution of 256^3. The loss weights are configured as $\lambda_1 = 1$, $\lambda_2 = 0.1$, $\lambda_3 = 0.1$, and $\lambda_4 = 10$. The entire training process for each shape is capped at approximately 3 minutes on an A100 GPU.

We evaluate on the GC-Object dataset [Sawdayee et al. 2023; Zhou et al. 2015], which contains 50 different types of common objects, and the Quadrupeds dataset [Tulsiani et al. 2017a; Xu et al. 2023], comprising 124 animal shapes. Both datasets were preprocessed following the SweepNet [Zhao et al. 2024] to ensure consistency.

4.2 Decomposition, Editing, and Parameter Efficiency

As shown in Fig. 1, 3D2EP effectively decomposes 3D shapes into extruded profile primitives, creating compact and interpretable representations. To evaluate editing capabilities, we imported the estimated path curves, scales, and profile curves into Blender, where users intuitively modified these parameters to create new shapes based on reference images. Editing tasks included adjustments to profile shapes, path curves, and scales. Fig. 4 highlights four groups of edited results from novice users who adapted decompositions to match reference images from the Objaverse dataset [Deitke et al. 2022]. All edits were completed in under 10 minutes, demonstrating the ease and efficiency of 3D2EP 's editing framework.

3D2EP also achieves significant parameter reduction compared to meshes and voxels while maintaining high reconstruction quality. As shown in Table 1, our method requires an average of just 372 parameters (0.075% of mesh parameters) for an IoU of 80%, compared to voxels requiring 10.838% with an average grid resolution of 33.92. Fig. 5 further shows that increasing parameter counts improves fidelity, while even low-parameter settings retain meaningful structures, enabling robust reconstructions and simplifying edits. The reduced parameter count streamlines workflows, allowing both novices and experts to efficiently perform high-level edits.

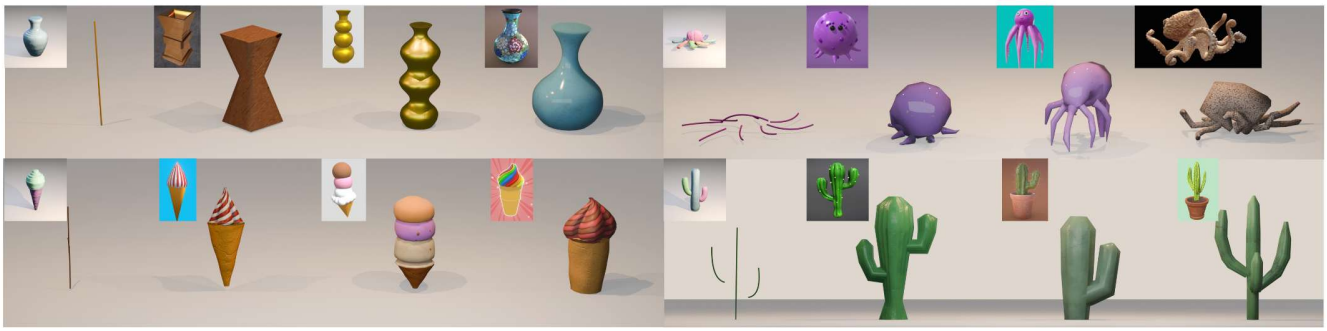


Fig. 4. **Shape editing results using our method integrated into Blender.** The path curves, profile curves, and scale parameters generated by our method can be imported into Blender, where identical functions enable shape editing. This allows novice users to intuitively modify these parameters and create new shapes based on a reference image. Each group consists of four results: the leftmost column shows the original shape and its path curve. The remaining three columns show edited shapes derived from the original, modified to align with the reference images (small insets).

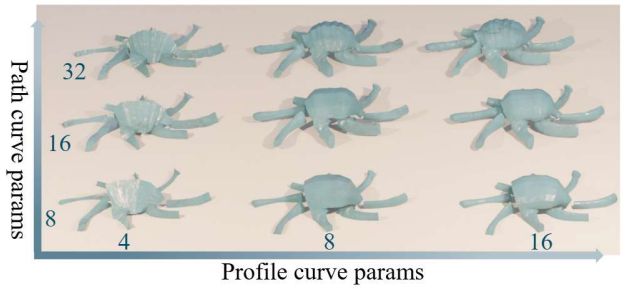


Fig. 5. **Visualization of 3D shape reconstructions with varying profile and path curve parameters.** The horizontal axis represents profile parameters (defining cross-sectional surfaces), and the vertical axis represents path parameters (defining extrusion directions). 3D2EP achieves reasonable reconstruction fidelity using fewer parameters, demonstrating efficiency.

Table 1. **Parameter efficiency comparison of 3D shape representations.** "Mesh" uses the most parameters, "Voxel" reduces them with a grid-based approach, and "3D2EP (ours)" achieves the highest efficiency with minimal path, profile, and primitive parameters.

Methods	Parameter Number	Percent
Mesh	492,812.64 *(Faces: 109,512.10, Vertices: 54,758.78)*	100.00%
Voxel	53,411.84 *(Grid Resolution: 33.92)*	10.84%
Ours	371.76 *(Path: 11.84, Profile: 16.00, Primitives: 4.54)*	0.075%

4.3 Results

Experimental Setup. We compare 3D2EP to superquadrics (SQ) [Paschalidou et al. 2019], cuboids (Cuboid) [Yang and Chen 2021], SECAD-Net [Li et al. 2023a], SHRED [Jones et al. 2022] and SweepNet [Zhao et al. 2024], on the GC-Object and Quadrupeds datasets, using the experimental settings from SweepNet.

Table 2. **Quantitative evaluations on GC-Object and Quadrupeds datasets.** Metrics include IoU (higher is better), Chamfer Distance (CD, lower is better), and Overlap (lower is better). Our method achieves superior reconstruction quality and compactness compared to all methods.

Dataset	Metric	Cuboid	SQ	SECAD	SweepNet	Ours
GC-Object	IoU ↑	0.797	0.742	0.654	0.829	**0.869**
	CD ↓	0.0183	0.0242	0.0282	0.0166	**0.0126**
	Overlap ↓	**0.0023**	0.3623	0.1984	0.0384	0.0143
Quadrupeds	IoU ↑	0.790	0.682	0.568	0.784	**0.833**
	CD ↓	0.0181	0.0313	0.0321	0.0184	**0.0152**
	Overlap ↓	**0.0019**	0.4093	0.1783	0.0462	0.0135

Quantitative Comparisons. In this paper, we use three evaluation metrics: Intersection over Union (IoU), Chamfer Distance (CD), and Overlap. (1) IoU measures voxel-level agreement between predicted and ground-truth shapes; higher values indicate better volumetric alignment. (2) Chamfer Distance computes the average bidirectional distance between surface points; lower values reflect more accurate surface geometry. (3) Overlap quantifies the extent of redundant coverage among primitives in the predicted shape. We first define the total number of overlaps, $\mathcal{N}_{\text{overlap}}$, using the same formulation as $\mathcal{L}_{\text{overlap}}$ in Eq. (19). To normalize, we divide $\mathcal{N}_{\text{overlap}}$ by the total number of occupied voxel cells in the final prediction, denoted as $\mathcal{N}_{\text{occ}}$. Here, an occupied voxel cell refers to a voxel grid location that is covered by at least one primitive with non-zero occupancy. The reported Overlap score is computed as Overlap $= \mathcal{N}_{\text{overlap}}/\mathcal{N}_{\text{occ}}$, where lower values indicate cleaner primitive separation and minimal redundancy. These metrics jointly evaluate reconstruction accuracy, geometric fidelity, and part-level separation. The evaluation results are summarized in Table 2. Reconstructions from Cuboid, Superquadric (SQ), and SECAD exhibit limited fidelity due to the restricted expressiveness of their primitive types, resulting in poor IoU and CD scores. While Cuboid achieves a low overlap ratio, this is not meaningful, as its reconstructions are overly coarse and fail to capture curved or articulated regions, leading to significant loss of detail. Superquadrics and SECAD, although slightly more flexible, still struggle with irregular structures and require more primitives

Table 3. **Robustness Evaluation to Segmentation Quality.** Comparison of Intersection over Union (IoU), Chamfer Distance (CD), and mask overlap metrics under varying levels of noise injection.

Setup	IoU↑	CD↓	Overlap↓
w/o noise	**0.869**	**0.0126**	0.0143
noisy prompting	0.858	0.0131	**0.0135**
view perturb	0.849	0.0144	0.0155
mask merge	0.847	0.0152	0.0328

Table 4. **Ablation Study.** Comparison of Intersection over Union (IoU), Chamfer Distance (CD), and overlap metrics across different variants. The full method (Ours) achieves the best performance, indicating the effect of incorporating axis, overlap, and mask loss terms.

Setup	IoU↑	CD↓	Overlap↓
w/o axis loss	0.853	0.0133	**0.0135**
w/o overlap loss	0.851	0.0158	0.3644
w/o mask loss	0.846	0.0142	0.0204
Ours	**0.869**	**0.0126**	0.0143

to approximate complex surfaces, reducing interpretability. SHRED produces fine-grained segmentations but operates directly on the input point cloud without reconstructing surfaces. Consequently, IoU, CD, and Overlap metrics are not applicable. Furthermore, SHRED outputs unstructured point clouds without explicit primitive representations, offering no support for parametric editing or consistent semantic decomposition. For these reasons, it is excluded from the user study. Futhermore, SweepNet better captures structural abstraction by combining 2D profiles and path curves, but it frequently suffers from overlapping primitives and inconsistent part coverage. These issues introduce redundancy and visual artifacts that hinder downstream tasks like editing or animation. In contrast, 3D2EP achieves high-fidelity reconstructions. It accurately captures fine details, articulated parts, and varying surface curvatures across diverse object categories. Our decomposition maintains a clean structural hierarchy, with minimal overlap and strong alignment with semantic part boundaries. This balance of geometric accuracy, interpretability, and editability makes our method particularly suitable for part-aware modeling and interactive applications.

Qualitative Comparisons. The qualitative results in Fig. 6 demonstrate that 3D2EP produces cohesive, compact, and smooth representations of complex 3D shapes, outperforming baseline approaches. By leveraging extruded profile primitives, 3D2EP captures intricate details, such as limbs and elongated features, in a natural and streamlined manner. Baseline methods often fall short in these areas: SECAD produces fragmented shapes, cuboid-based methods result in rigid, overly segmented outputs, and superquadrics require numerous components to approximate curves, leading to inconsistencies. SweepNet tends to produce overlapping or fragmented primitives, which can make the decomposition less interpretable and harder to edit. SHRED achieves detailed surface reconstruction through local patch prediction, but the output consists of numerous unstructured point cloud patches, limiting explicit control and semantic clarity. In contrast, 3D2EP generates smooth, continuous surfaces and interpretable segments that naturally align with organic shapes, such as animal bodies and jointed forms.

Table 5. **User Study: Percentage of votes received by each method for reconstruction quality and editing ability.** Our method received the highest percentage compared with other baselines.

Methods	Cuboid	SQ	SECAD	SweepNet	Ours
Vote %	0%	0%	0%	12.375%	87.625%

Robustness Evaluation to Segmentation Quality. To assess the robustness of 3D2EP to segmentation quality, we inject controlled noise into the 2D masks using three strategies: (1) **Noisy Prompting**: for each view, part masks are independently selected with a probability of 0.15, and their positive points are randomly shifted while negative points are sampled near object boundaries to perturb the SAM prompts, generating noisy masks; (2) **Multiview Perturbation**: one to three views are randomly selected, and in each selected view, part masks are independently selected with a probability of 0.15 to exchange their mask indices, introducing label inconsistencies; the selected masks are also slightly eroded or dilated using a 3×3 kernel to perturb boundaries; (3) **Mask Merging**: pairs of part masks are randomly merged across all views to simulate structural over-segmentation errors. Detailed descriptions of these procedures are provided in the supplementary material.

Quantitative results in Table 3 show that Intersection over Union (IoU) and Chamfer Distance (CD) remain stable across different noise types, indicating that decomposition quality is largely unaffected by noisy prompts or multiview inconsistencies. Only mask merging leads to a notable increase in overlap, reflecting the undesired combination of distinct parts. Nevertheless, since our framework supports interactive refinement through user prompts before decomposition, such errors can be easily corrected. This approach allowing users full control over the process. As shown in Fig. 7, visual results further confirm that our method consolidates corrupted masks and additional regions into stable 3D decompositions, while merged masks result in different part structures. More visual results can be found in supplementary material.

Ablation Study. We present an ablation study to evaluate the impact of removing the axis, overlap, and mask loss terms, in Table 4 and Fig. 8. Removing the axis loss results in twisted, non-smooth curves at endpoints, reducing their utility. Eliminating the overlap loss causes significant overlaps between primitives, compromising geometric fidelity. Omitting the mask loss leads to decompositions losing semantic coherence, producing less meaningful splits. In contrast, the full method achieves smooth curves, minimal overlaps, and coherent decompositions. This is reflected in the quantitative ablation results, which shows that the overlap loss has the most substantial effect on most metrics. This shows that all loss terms are essential for accurate and interpretable decompositions.

User Study. We conducted a user study using GC-Object. Sixteen comparison pairs were randomly sampled, with 50 participants per pair evaluating reconstruction quality and editing ability. Each

comparison included rendered reconstructions and primitive visualizations from different methods. Table 5 shows that our method received the greatest share of the votes by a considerable margin. SweepNet received 12.375%, while methods such as Cuboid, SQ, and SECAD receives no votes. This highlights the superiority of 3D2EP in delivering high-quality reconstructions and editing capabilities.

5 Discussion and Conclusion

Limitations. The proposed primitive representation struggles with parts where the profile twists along the curve (e.g., ropes) or amorphous structures lacking a coherent cross-section (e.g., draped cloth), where no dominant extrusion direction can be defined. It is also less effective when segmentation masks are under-segmented, as merged parts can hinder meaningful primitive fitting, leading to inconsistent decompositions. User-guided corrections before decomposition can mitigate these issues.

Conclusion. This paper introduces a novel approach to 3D shape decomposition, offering a compact, interpretable, and efficient representation that facilitates editing, redesign, and compression. By representing shapes as a collection of differentiable, parametric primitives, 3D2EP achieves precise reconstruction of intricate geometries while maintaining simplicity for intuitive part-level editing. 3D2EP fits primitives directly to each input 3D shape with only 2D part segmentation masks as supervision. Extensive evaluation demonstrates the effectiveness of 3D2EP compared to existing methods. This work provides a versatile tool for 3D design and opens new avenues for applications in shape editing and compression.

References

Alan H Barr. 1981. Superquadrics and angle-preserving transformations. *IEEE Computer graphics and Applications* 1, 01 (1981), 11–23.

Jonathan T Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P Srinivasan. 2021. Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5855–5864.

Jonathan T Barron, Ben Mildenhall, Dor Verbin, Pratul P Srinivasan, and Peter Hedman. 2023. Zip-nerf: Anti-aliased grid-based neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 19697–19705.

Alexandre Boulch and Renaud Marlet. 2022. Poco: Point convolution for surface reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6302–6314.

Rohan Chabra, Jan E Lenssen, Eddy Ilg, Tanner Schmidt, Julian Straub, Steven Lovegrove, and Richard Newcombe. 2020. Deep local shapes: Learning local sdf priors for detailed 3d reconstruction. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXIX 16*. Springer, 608–625.

Zhiqin Chen and Hao Zhang. 2019. Learning implicit fields for generative shape modeling. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5939–5948.

Julian Chibane, Thiemo Alldieck, and Gerard Pons-Moll. 2020. Implicit functions in feature space for 3D shape reconstruction and completion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6970–6981.

Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. 2022. Objaverse: A Universe of Annotated 3D Objects. *arXiv preprint arXiv:2212.08051* (2022).

Boyang Deng, Kyle Genova, Soroosh Yazdani, Sofien Bouaziz, Geoffrey Hinton, and Andrea Tagliasacchi. 2020. Cvxnet: Learnable convex decomposition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 31–44.

Kyle Genova, Forrester Cole, Daniel Vlasic, Aaron Sarna, William T Freeman, and Thomas Funkhouser. 2019. Learning shape templates with structured implicit functions. In *Proceedings of the IEEE/CVF international conference on computer vision*. 7154–7164.

Simon Giebenhain and Bastian Goldlücke. 2021. Air-nets: An attention-based framework for locally conditioned implicit representations. In *2021 International Conference on 3D Vision (3DV)*. IEEE, 1054–1064.

Junlin Han, Filippos Kokkinos, and Philip Torr. 2025. Vfusion3d: Learning scalable 3d generative models from video diffusion models. In *European Conference on Computer Vision*. Springer, 333–350.

Junlin Han, Jianyuan Wang, Andrea Vedaldi, Philip Torr, and Filippos Kokkinos. 2024. Flex3d: Feed-forward 3d generation with flexible reconstruction model and input view curation. *arXiv preprint arXiv:2410.00890* (2024).

Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. 2023. Lrm: Large reconstruction model for single image to 3d. *arXiv preprint arXiv:2311.04400* (2023).

Jiahui Huang, Zan Gojcic, Matan Atzmon, Or Litany, Sanja Fidler, and Francis Williams. 2023. Neural kernel surface reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4369–4379.

Chiyu Jiang, Avneesh Sud, Ameesh Makadia, Jingwei Huang, Matthias Nießner, Thomas Funkhouser, et al. 2020. Local implicit grid representations for 3d scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6001–6010.

R Kenny Jones, Aalia Habib, and Daniel Ritchie. 2022. Shred: 3d shape region decomposition with learned local operations. *ACM Transactions on Graphics (TOG)* 41, 6 (2022), 1–11.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. 2023. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4015–4026.

Florian Kluger, Hanno Ackermann, Eric Brachmann, Michael Ying Yang, and Bodo Rosenhahn. 2021. Cuboids revisited: Learning robust 3d shape fitting to single rgb images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13070–13079.

Lingxiao Li, Minhyuk Sung, Anastasia Dubrovina, Li Yi, and Leonidas J Guibas. 2019. Supervised fitting of geometric primitives to 3d point clouds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2652–2660.

Manyi Li and Hao Zhang. 2021. D2im-net: Learning detail disentangled implicit fields from single images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10246–10255.

Pu Li, Jianwei Guo, Xiaopeng Zhang, and Dong-Ming Yan. 2023a. Secad-net: Self-supervised cad reconstruction by learning sketch-extrude operations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16816–16826.

Runjia Li, Junlin Han, Luke Melas-Kyriazi, Chunyi Sun, Zhaochong An, Zhongrui Gui, Shuyang Sun, Philip Torr, and Tomas Jakab. 2024. DreamBeast: Distilling 3D Fantastical Animals with Part-Aware Knowledge Transfer. *arXiv preprint arXiv:2409.08271* (2024).

Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H Taylor, Mathias Unberath, Ming-Yu Liu, and Chen-Hsuan Lin. 2023b. Neuralangelo: High-fidelity neural surface reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8456–8465.

Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. 2023. Magic3d: High-resolution text-to-3d content creation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 300–309.

Yujia Liu, Anton Obukhov, Jan Dirk Wegner, and Konrad Schindler. 2024. Point2CAD: Reverse Engineering CAD Models from 3D Point Clouds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3763–3772.

Xiaoyang Lyu, Yang-Tian Sun, Yi-Hua Huang, Xiuzhe Wu, Ziyi Yang, Yilun Chen, Jiangmiao Pang, and Xiaojuan Qi. 2024. 3dgsr: Implicit surface reconstruction with 3d gaussian splatting. *ACM Transactions on Graphics (TOG)* 43, 6 (2024), 1–12.

Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.

Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. Instant neural graphics primitives with a multiresolution hash encoding. *ACM transactions on graphics (TOG)* 41, 4 (2022), 1–15.

Chengjie Niu, Jun Li, and Kai Xu. 2018. Im2struct: Recovering 3d shape structure from a single rgb image. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4521–4529.

Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. 2019. Deepsdf: Learning continuous signed distance functions for shape representation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 165–174.

Despoina Paschalidou, Luc Van Gool, and Andreas Geiger. 2020. Learning unsupervised hierarchical part decomposition of 3d objects from a single rgb image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1060–1070.

Despoina Paschalidou, Ali Osman Ulusoy, and Andreas Geiger. 2019. Superquadrics revisited: Learning 3d shape parsing beyond cuboids. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10344–10353.

Songyou Peng, Michael Niemeyer, Lars Mescheder, Marc Pollefeys, and Andreas Geiger. 2020. Convolutional occupancy networks. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part III 16*. Springer, 523–540.

Ben Poole, Ajay Jain, Jonathan T Barron, and Ben Mildenhall. 2022. DreamFusion: Text-to-3D using 2D Diffusion. In *The Eleventh International Conference on Learning Representations*.

Minghan Qin, Wanhua Li, Jiawei Zhou, Haoqian Wang, and Hanspeter Pfister. 2024. Langsplat: 3d language gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20051–20060.

Amit Raj, Srinivas Kaza, Ben Poole, Michael Niemeyer, Nataniel Ruiz, Ben Mildenhall, Shiran Zada, Kfir Aberman, Michael Rubinstein, Jonathan Barron, et al. 2023. Dreambooth3d: Subject-driven text-to-3d generation. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2349–2359.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. 2024. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714* (2024).

Shunsuke Saito, Zeng Huang, Ryota Natsume, Shigeo Morishima, Angjoo Kanazawa, and Hao Li. 2019. Pifu: Pixel-aligned implicit function for high-resolution clothed human digitization. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2304–2314.

Haim Sawdayee, Amir Vaxman, and Amit H Bermano. 2023. Orex: Object reconstruction from planar cross-sections using neural fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20854–20862.

Gopal Sharma, Difan Liu, Subhransu Maji, Evangelos Kalogerakis, Siddhartha Chaudhuri, and Radomír Měch. 2020. Parsenet: A parametric surface fitting network for 3d point clouds. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VII 16*. Springer, 261–276.

Dmitriy Smirnov, Matthew Fisher, Vladimir G Kim, Richard Zhang, and Justin Solomon. 2020. Deep parametric shape predictions using distance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 561–570.

Chunyi Sun, Junlin Han, Weijian Deng, Xinlong Wang, Zishan Qin, and Stephen Gould. 2023. 3d-gpt: Procedural 3d modeling with large language models. *arXiv preprint arXiv:2310.12945* (2023).

Chun-Yu Sun, Qian-Fang Zou, Xin Tong, and Yang Liu. 2019. Learning adaptive hierarchical cuboid abstractions of 3d shape collections. *ACM Transactions on Graphics (TOG)* 38, 6 (2019), 1–13.

Andrea Tagliasacchi, Ibraheem Alhashim, Matt Olson, and Hao Zhang. 2012. Mean curvature skeletons. In *Computer Graphics Forum*, Vol. 31. Wiley Online Library, 1735–1744.

Jiapeng Tang, Jiabao Lei, Dan Xu, Feiying Ma, Kui Jia, and Lei Zhang. 2021. Sa-convonet: Sign-agnostic optimization of convolutional occupancy networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 6504–6513.

Wenwen Tong, Chonghao Sima, Tai Wang, Li Chen, Silei Wu, Hanming Deng, Yi Gu, Lewei Lu, Ping Luo, Dahua Lin, et al. 2023. Scene as occupancy. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 8406–8415.

Shubham Tulsiani, Hao Su, Leonidas J Guibas, Alexei A Efros, and Jitendra Malik. 2017a. Learning shape abstractions by assembling volumetric primitives. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2635–2643.

Shubham Tulsiani, Hao Su, Leonidas J Guibas, Alexei A Efros, and Jitendra Malik. 2017b. Learning shape abstractions by assembling volumetric primitives. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 2635–2643.

Dor Verbin, Peter Hedman, Ben Mildenhall, Todd Zickler, Jonathan T Barron, and Pratul P Srinivasan. 2022. Ref-nerf: Structured view-dependent appearance for neural radiance fields. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 5481–5490.

Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. 2021. NeuS: Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction. *NeurIPS* (2021).

Yiming Wang, Qin Han, Marc Habermann, Kostas Daniilidis, Christian Theobalt, and Lingjie Liu. 2023. NeuS2: Fast Learning of Neural Implicit Surfaces for Multi-view Reconstruction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3295–3306.

Francis Williams, Zan Gojcic, Sameh Khamis, Denis Zorin, Joan Bruna, Sanja Fidler, and Or Litany. 2022. Neural fields as learnable kernels for 3d reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18500–18510.

Jiacong Xu, Yi Zhang, Jiawei Peng, Wufei Ma, Artur Jesslen, Pengliang Ji, Qixin Hu, Jiehua Zhang, Qihao Liu, Jiahao Wang, et al. 2023. Animal3d: A comprehensive dataset of 3d animal pose and shape. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 9099–9109.

Siming Yan, Zhenpei Yang, Chongyang Ma, Haibin Huang, Etienne Vouga, and Qixing Huang. 2021. Hpnet: Deep primitive segmentation using hybrid representations. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2753–2762.

Kaizhi Yang and Xuejin Chen. 2021. Unsupervised learning for cuboid shape abstraction via joint segmentation from point clouds. *ACM Transactions On Graphics (TOG)* 40, 4 (2021), 1–11.

Kai Zhang, Gernot Riegler, Noah Snavely, and Vladlen Koltun. 2020. Nerf++: Analyzing and improving neural radiance fields. *arXiv preprint arXiv:2010.07492* (2020).

Zijun Zhang. 2018. Improved adam optimizer for deep neural networks. In *2018 IEEE/ACM 26th international symposium on quality of service (IWQoS)*. Ieee, 1–2.

Mingrui Zhao, Yizhi Wang, Fenggen Yu, Changqing Zou, and Ali Mahdavi-Amiri. 2024. SweepNet: Unsupervised Learning Shape Abstraction via Neural Sweepers. In *European Conference on Computer Vision*. Springer.

Yang Zhou, Kangxue Yin, Hui Huang, Hao Zhang, Minglun Gong, and Daniel Cohen-Or. 2015. Generalized cylinder decomposition. *ACM Trans. Graph.* 34, 6 (2015), 171–1.

Chuhang Zou, Ersin Yumer, Jimei Yang, Duygu Ceylan, and Derek Hoiem. 2017. 3d-prnn: Generating shape primitives with recurrent neural networks. In *Proceedings of the IEEE International Conference on Computer Vision*. 900–909.

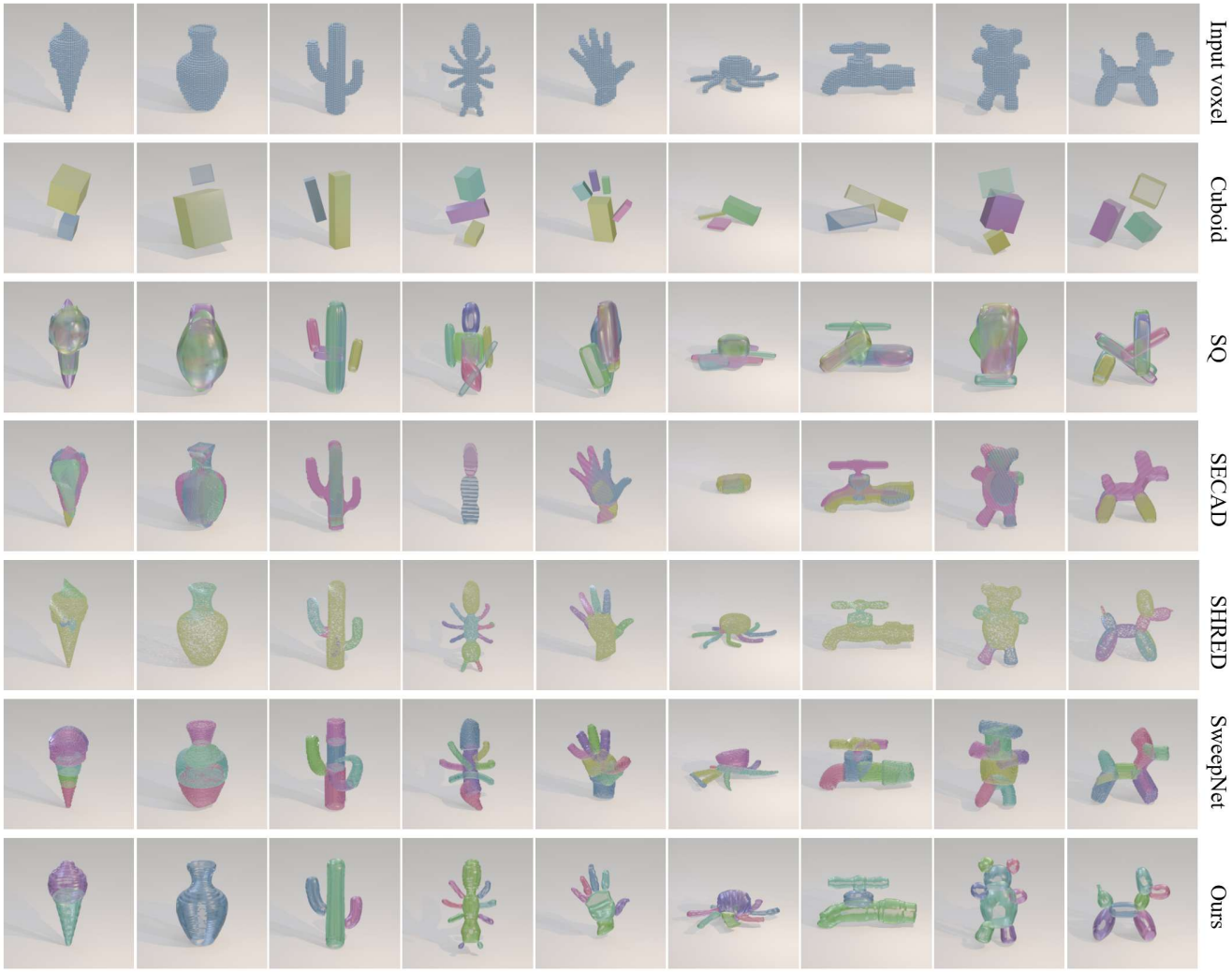


Fig. 6. **Qualitative visual comparison on GC-Object and Quadrupeds.** Rows represent results from different methods: the input voxel, Cuboid, SQ, SECAD, SHRED, SweepNet, and Ours. The visualizations show each method's ability to reconstruct and represent complex 3D shapes. Our method combines structural preservation and compact representation, producing results that align closely with the original shapes while maintaining simplicity and interpretability.

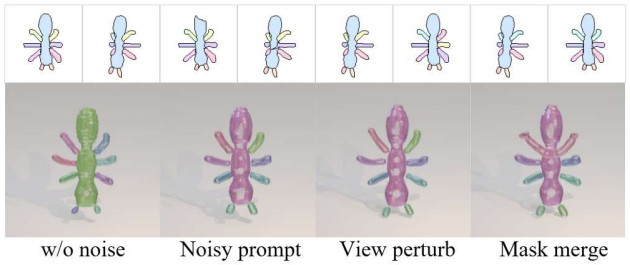


Fig. 7. **Robustness Evaluation to Segmentation Quality.** Top: segmentation masks from different views. Bottom: corresponding 3D reconstructions. Our method consolidates noisy prompts, view inconsistencies, and merged parts into coherent decompositions.

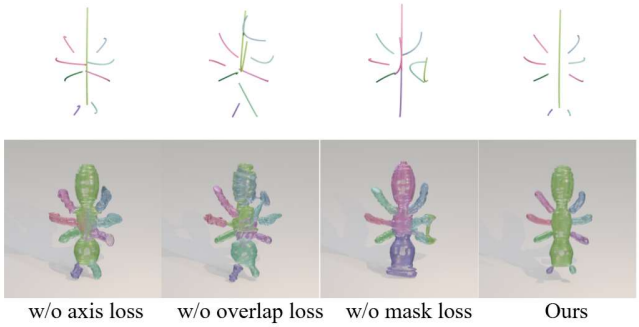


Fig. 8. **Ablation Study.** The top row shows path curves, while the bottom row presents 3D shapes. Removing the axis, overlap, or mask loss reduces alignment, compactness, and accuracy compared to the full method.